

Beam Current Limits

USPAS - Beam Physics with Intense Space Charge

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1. Beam current limits in axisymmetric focusing channels
2. Beam current limits in quadrupolar focusing channels

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1.1 Radial envelope equations

For axially symmetric focusing systems, we derived the radial envelope equation:

$$\tilde{r}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{r}' + \left(\frac{\gamma''}{2\gamma\beta^2}\right)\tilde{r} + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2\tilde{r} - \left(\frac{\langle p_\theta \rangle}{\gamma m\beta c}\right)^2 \frac{1}{\tilde{r}^3} - \frac{\varepsilon_r^2}{\tilde{r}^3} - \frac{Q}{2\tilde{r}} = 0 \quad (1)$$

$$p_\theta \equiv \gamma m r^2 \dot{\theta} + q r A_\theta, \quad (2)$$

$$\varepsilon_r \equiv \sqrt{\langle r^2 \rangle \langle r'^2 \rangle - \langle r r' \rangle^2 + \langle r^2 \rangle \langle r^2 \theta'^2 \rangle - \langle r^2 \theta' \rangle^2} \quad (3)$$

$$Q \equiv \frac{1}{mc^2 \beta^2 \gamma^3} \frac{q \langle \lambda(r) \rangle}{2\pi \epsilon_0} \quad (4)$$

1.2 Solenoid focusing

[Image of solenoid]

1.3 Current limit in solenoid

- Assume infinitely long solenoid: $\mathbf{B} = B_z \hat{z}$.
- Assume $\gamma' = \gamma'' = 0$.
- Assume $p_\theta = 0$ and $\epsilon_r = 0$ to minimizing “repelling” forces.

$$\tilde{r}'' + \left(\frac{\omega_c}{2\gamma\beta c} \right)^2 \tilde{r} - \frac{Q}{2\tilde{r}} = 0 \quad (5)$$

At equilibrium, $\tilde{r}' = \tilde{r}'' = 0$:

$$Q = \frac{1}{2} \left(\frac{\omega_c \tilde{r}}{\gamma\beta c} \right)^2 \quad (6)$$

1.4 Low-energy current limit derived from force balance

- Particle moves with angular velocity $\theta = \omega r$.
- Particle sees linear focusing force from solenoid field $B_z = \omega_c m/q$.
- Within beam envelope, particle sees linear defocusing force from space charge.

$$\underbrace{m\omega^2 r}_{\text{centrifugal}} + \underbrace{\frac{Qm\beta^2 c^2}{2\tilde{r}^2} r}_{\text{centrifugal}} = \underbrace{qv_\theta B_z}_{\text{magnetic}} \quad (7)$$

$$\omega^2 r + \frac{Q\beta^2 c^2}{2\tilde{r}^2} r = \omega_c \omega r \quad (8)$$

$$Q(\omega) = \frac{2\tilde{r}^2}{\beta^2 c^2} (\omega\omega_c - \omega^2) \quad (9)$$

$$\omega^* = \max_{\omega} Q(\omega) = \frac{\omega_c}{2} \quad (10)$$

$$Q(\omega^*) = \frac{1}{2} \left(\frac{\omega_c \tilde{r}}{\beta c} \right)^2 \quad (11)$$

Equivalent to result from envelope equations when $\gamma^2 \rightarrow 1$:

$$Q = \frac{1}{2} \left(\frac{\omega_c \tilde{r}}{\gamma \beta c} \right)^2 \quad (12)$$

1.5 Einzel lens

[Image of Einzel lens]

Image of beam envelope]

Ignore centrifugal term, emittance term, and solenoid term (B_z) in radial envelope equation.

$$\tilde{r}'' + \frac{\gamma'}{\gamma\beta^2}\tilde{r}' + \frac{\gamma''}{2\gamma\beta^2} - \frac{Q}{2\tilde{r}} = 0 \quad (13)$$

Assume non-relativistic beam: $\beta \ll 1$, $\gamma \approx 1$:

$$\gamma' = \beta\beta'\gamma^3 \approx \beta\beta' \quad (14)$$

$$\gamma'' \approx \beta'\beta' + \beta\beta'' \quad (15)$$

$$\tilde{r}'' + \frac{\beta'}{\beta}\tilde{r}' + \frac{1}{2} \left[\frac{\beta'^2}{\beta^2} + \frac{\beta''}{\beta} \right] \tilde{r} - \frac{Q}{2\tilde{r}} = 0 \quad (16)$$

Define new variable R :

$$\tilde{r} = \left(\frac{\beta_0}{\beta} \right)^{1/2} R, \quad (17)$$

where β_0 is the velocity at $s = 0$.

$$\tilde{r}' = \left(\frac{\beta}{\beta_0} \right)^{-1/2} R' - \frac{1}{2} \left(\frac{\beta}{\beta_0} \right)^{-3/2} \left(\frac{\beta'}{\beta} \right) R \quad (18)$$

$$\tilde{r}'' = \left(\frac{\beta}{\beta_0} \right)^{-1/2} R'' - \left(\frac{\beta}{\beta_0} \right)^{-3/2} \left(\frac{\beta'}{\beta_0} \right) R' + \frac{3}{4} \left(\frac{\beta}{\beta_0} \right)^{-5/2} \left(\frac{\beta'}{\beta_0} \right)^2 R - \frac{1}{2} \left(\frac{\beta}{\beta_0} \right)^{-3/2} \left(\frac{\beta''}{\beta} \right) R \quad (19)$$

$$\boxed{R'' + \frac{3}{4} \left(\frac{\beta}{\beta} \right)^2 R - \frac{1}{2} \frac{\beta}{\beta_0} \frac{Q}{R} = 0.} \quad (20)$$

Approximate potential as $\phi(r, z) = \phi_0 \cos(\pi z/L)$.d

Conservation of energy:

$$\frac{1}{2}mv^2 + 2q\phi = \text{constant} \quad (21)$$

$$\Delta(v^2) = -\Delta\left(\frac{2q\phi}{m}\right) \quad (22)$$

$$v^2 - v_0^2 = -\frac{2q\phi_0}{m} \cos\left(\frac{\pi z}{L}\right) \quad (23)$$

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2.1 RMS envelope equations

$$\tilde{x}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{x}' + \kappa_x(s)\tilde{x} - \frac{\varepsilon_x(s)^2}{\tilde{x}^3} - \frac{2Q}{\tilde{x} + \tilde{y}} = 0, \quad (24)$$

$$\tilde{y}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{y}' + \kappa_y(s)\tilde{y} - \frac{\varepsilon_y(s)^2}{\tilde{y}^3} - \frac{2Q}{\tilde{x} + \tilde{y}} = 0. \quad (25)$$